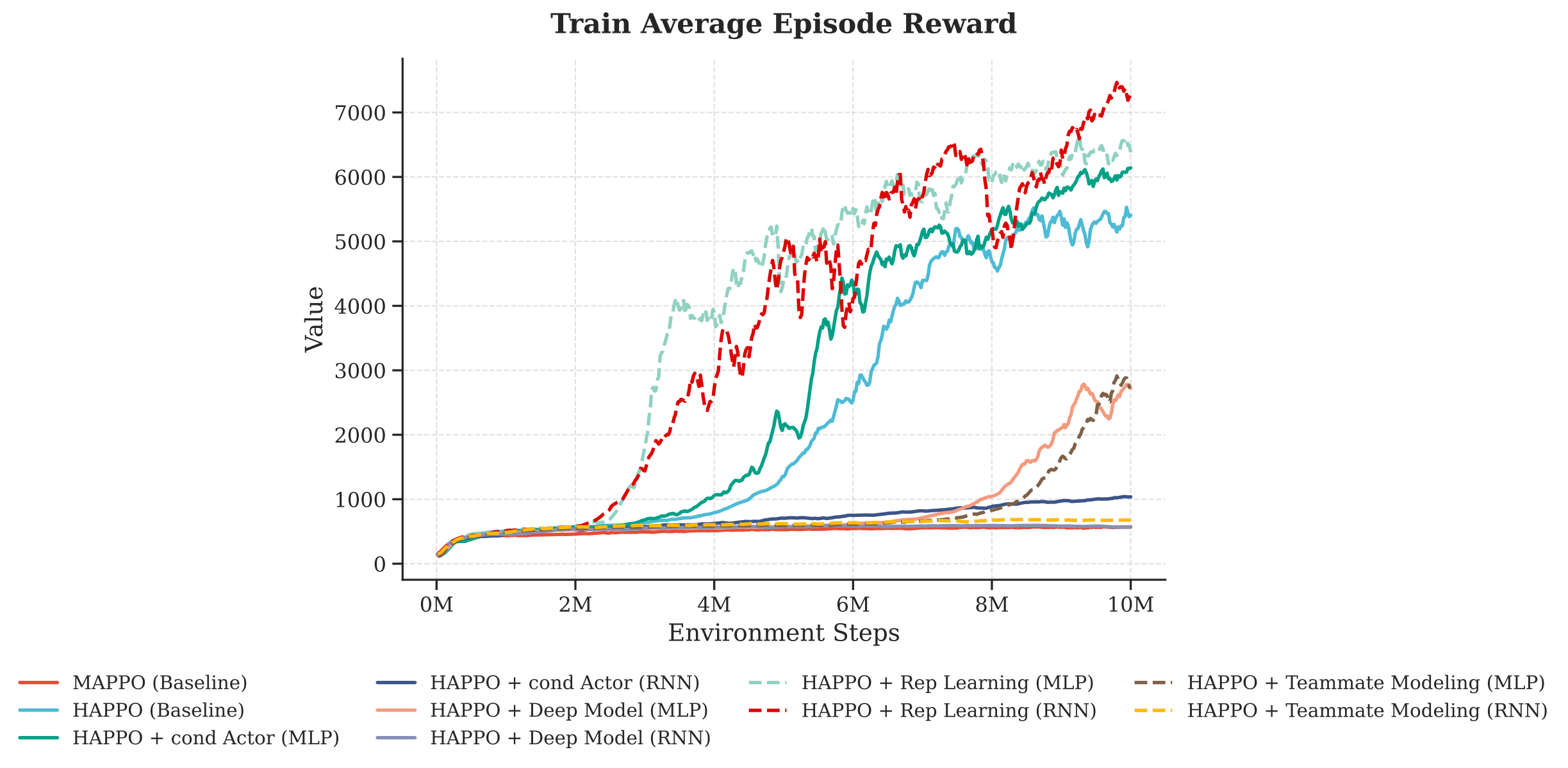

Train Average Episode Reward
Value
7000
6000
5000
4000
3000
2000
1000
0
0M
2M
4M
6M
8M
10M
Environment Steps
MAPPO (Baseline)
HAPPO (Baseline)
HAPPO + cond Actor (MLP)
HAPPO + cond Actor (RNN)
HAPPO + Deep Model (MLP)
HAPPO + Deep Model (RNN)
HAPPO + Rep Learning (MLP)
HAPPO + Rep Learning (RNN)
HAPPO + Teammate Modeling (MLP)
HAPPO + Teammate Modeling (RNN)